Which Damage Do the Images Support? Evidence-Bounded 3D Damage Estimation on Turbine Blades

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Abstract

A dark patch in a photograph of an aero-engine blade may be a dent, a stain, a lighting change or a small camera error, and an inspector needs to know which, where, and how deep. We study evidence-bounded 3D damage estimation: given a healthy CAD model and a few calibrated images, report a displacement field on the blade surface only when the images support a physical change better than every non-physical explanation, and otherwise state that the evidence is insufficient. Our estimator compares a nuisance-only explanation (per-view pose, light gains, an albedo patch) with the same explanation plus a continuous dent through a differentiable renderer, proposes dents with a Rao score test over the whole surface, and thresholds the fit gain at a split-conformal level calibrated on healthy and stain-only blades. On a sealed test of 30 unseen blades from BladeDelta, a benchmark of 696 rendered cases on public Rotor37 geometry plus 456 cases on six industrial blade designs, the method reports 12 of 13 visible dents deeper than 0.2 mm at the correct location (median depth error 0.03 mm) and declines most dents below 0.1 mm; at a matched 5% false-alarm rate it locates 51% of visible dents against at most 19% for CAD-prior deformation fields and a lifted 2D detector, and a second sealed test reproduces the result. On the industrial designs, the frozen model raises false alarms on 67% of healthy or stain-only blades; we trace this to a silhouette pose bias and a reference correction that does not move with the camera, and correcting both, without retraining, lowers false alarms on held-out blades from 51% to 4% (AUROC 0.72 to 0.95) at no cost on Rotor37. The result is a damage estimator whose claims come with a calibrated statement of when the photographs do not support them.

1. Introduction

Turbine and compressor blades are inspected with cameras, and the inspector's question is physical: is there a dent, where on the blade is it, and how deep is it? Damage of interest is small. On a blade imaged at 512×512 pixels, a dent of 0.07 mm changes a few hundred pixels by two to seven grey levels. A one-degree camera error, a ten-percent lighting change or a stain changes thousands.

Current vision methods answer an easier question. Industrial anomaly detection localises unusual pixels [2, 21], including under pose and illumination change [4, 14]. Reference-based methods compare a photograph with a CAD model and output boxes [3], or lift per-view scores into an anomaly volume [8]. CAD-conditioned reconstruction fits a deformation field to sparse posed views [20, 24], and leaves measurement-quality damaged surfaces to future work [20]. These outputs do not state whether the images support a physical change rather than a nuisance explanation.

Our own earlier experiments made this concrete. A render-and-compare fit on public Rotor37 blade geometry produced false dents on 14 of 14 healthy or colour-only blades under a one-degree camera error, and on unfamiliar dents the best image match selected a worse 3D shape in 8 of 12 depth sweeps. Better image agreement is not better geometry, and a plausible reconstruction can choose the wrong physical explanation.

We therefore formulate evidence-bounded 3D damage estimation. The input is a healthy CAD mesh and K calibrated images; the output is a per-vertex displacement field together with a decision, supported or not supported by these views. Our estimator treats this as a nested hypothesis comparison. H_0 explains the images with nuisance only: per-view pose corrections, light gains and a surface albedo patch. H_1 adds a continuous dent. The fit gain $L(H_0) - L(H_1)$ is thresholded at a value calibrated on negatives, giving a controlled false-change rate.

Making this work required separating the roles of the evidence. Pose is fitted from silhouettes, which do not depend on lighting; the dent is compared against nuisance only on interior surface pixels that are not grazing, not near a silhouette and not on the blade root. Dent hypotheses come from a Rao score test evaluated for every surface location and radius with one backward pass through the renderer, and the dent and colour-patch explanations are searched symmetrically, so a colour patch fitted first cannot hide a dent. We also report observability: on our development split, 70% of dents placed on camera-facing surfaces are less than half visible in the two input views, and the correct output for those is to abstain.

Contributions:

- The task of evidence-bounded 3D damage estimation from sparse views and healthy CAD, with metrics that separate localisation, shape accuracy, false physical change and abstention on unobservable damage.
- An estimator combining a differentiable renderer with a neural shader, silhouette-only pose, interior-domain nested hypothesis testing, score-test proposals and split-conformal calibration.
- BladeDelta, a benchmark built on public Rotor37 geometry [5] with exact damaged meshes, colour-only negatives, three dent families that differ from the estimator's own profile, and controlled camera and lighting perturbations, with a protocol and sealed test split fixed before evaluation.

2. Related Work

Industrial and 3D anomaly detection. Memory-bank methods such as PatchCore [21] localise anomalous pixels; MVTec 3D-AD [2] and Anomaly-ShapeNet [16] add measured or synthetic 3D. Pose-agnostic settings use multi-pose benchmarks [26] and Gaussian-splat references [14], and M^{2}AD [4] benchmarks view–illumination interplay. Multi-view settings include RAD [6] and SiM3D [8], which lifts per-view scores into an anomaly volume; recent work warns that adding views can leak information in multi-view evaluation [7]. Synthetic anomalies have been argued sufficient for 3D detection [23], and incomplete-modality methods train with 3D and test on RGB [22]. These methods output anomaly scores; they do not estimate a physical surface change or state whether the views support one.

Reference CAD and blade inspection. Looking 3D [3] aligns a query photograph with a reference shape and localises abnormal regions with boxes. BladeSynth [10] renders procedural nicks, dents, scratches and corrosion on aero-engine blades for 2D detection. Model-free universal detection reconstructs a normal shape by GAN inversion [18]. We instead output a 3D displacement field on a known design.

CAD-conditioned reconstruction. CAD-NeRF [24] retrieves a CAD model and jointly optimises deformation, pose and appearance from few uncalibrated views. CADSplat [20] fits a shared deformation field to sparse posed views with a CAD prior and explicitly leaves measurement-quality deformed surfaces to future work. Our free-form deformation baseline follows this family. The difference is the output: a decision on whether the deformation is supported, with a calibrated false-change rate.

Inverse rendering and uncertainty. Inverse rendering separates shape and material from images [19, 25]; we use differentiable rasterisation [15] with a learned shader for the same purpose on a known geometry. Radiance-field uncertainty [11, 13] and observation completeness [17] quantify where a reconstruction is unreliable. They do not test a physical-change hypothesis against nuisance explanations, which is the question an inspector needs answered.

Selective prediction and calibration. Abstaining when evidence is insufficient is the setting of selective classification, where a model may reject inputs to guarantee a target risk on the rest [12]. We calibrate the abstention threshold with split-conformal quantiles on negatives [1], which bounds the false-change rate under exchangeability; unlike classification, the quantity being accepted or rejected is a physical surface change, and the score is the evidence one hypothesis has over a nuisance-only explanation.

3. Method

Setup. We are given a healthy mesh $\mathcal{M} = (V, F)$ with vertices $V \in \mathbb{R}^{N\times 3}$, K observed images $I_1, \ldots, I_K$ with nominal intrinsics and extrinsics, and nominal lighting. We seek a displacement field $D \in \mathbb{R}^{N\times 3}$ on the airfoil and a decision $s \in \{\text{supported}, \text{not supported}\}$. Nuisance variables η are a per-view pose correction (a rotation about the part and a camera-frame translation), three light gains, and an optional albedo patch.

Differentiable renderer. We rasterise $\mathcal{M}+D$ with nvdiffrast [15] using smooth vertex normals. A neural shader maps the per-pixel world normal n, position x and view direction ω to one radiance component per

light, $R_k(n, x, \omega)$, and an environment term R_0. Because light transport is linear in each light's intensity, the pixel colour is $T\big(a \odot (R_0 + \sum_k g_k R_k)\big)$ for albedo multiplier a, light gains g and a learned monotone tone curve T, followed by anti-aliasing and the sensor's pixel filter. An unstructured shader that took g as an input learned to ignore it, and lighting changes then produced spurious dents. The shader is trained once, on renders of blade geometries disjoint from all evaluation geometries, with a paired loss on image differences between lighting settings. The healthy CAD rendered under the nominal capture provides a per-pixel correction $B = I^{\text{ref}} - \hat{I}(\mathcal{M}; \eta_0)$ that absorbs static shader error. On a held-out blade its mean error is 0.92 grey levels; image changes caused by a lighting change correlate with the reference renderer at 0.90–0.98, and changes caused by a dent at 0.96–0.98 in the 120° view.

Nuisance fit (H_0). Each view's pose is initialised by Euclidean ECC alignment of the observed and reference images, then refined by matching anti-aliased silhouettes to the observed foreground, coarse to fine. Silhouettes do not depend on lighting, which prevents the pose from compensating for a brightness change. With pose fixed, light gains are fitted on the full image.

Evidence domain. All further comparisons use a fixed interior domain Ω: airfoil pixels at least 6 px from any silhouette or self-occlusion edge, excluding the blade root and pixels where $|n \cdot \omega| < 0.35$. Both hypotheses are evaluated on the same Ω, so their losses are comparable.

Dent model (H_1). A dent is parameterised by a subvertex centre c (an anchor vertex plus a tangent offset), depth $\delta \geq 0$, radius r, aspect ratio, orientation, profile exponent p and a signed rim coefficient κ:

$$D_i = -\delta \left[(1 - s_i^2)_+^p - \tfrac{\kappa}{2} \rho(s_i) \right] n_c, \qquad (1)$$

where s_i is the anisotropic normalised distance of vertex i from c in the tangent plane, ρ a ring profile and n_c the centre normal, restricted to same-side airfoil vertices. None of the benchmark's truth families matches this profile exactly.

Score-test proposals. Let $h \in \mathbb{R}^N$ be an inward normal displacement. One backward pass gives $g = -\partial L_\Omega / \partial h$ at $h = 0$. For a kernel $w_{c,r}$ centred at c with radius r, the normalised score $\langle w_{c,r}, g \rangle / \|w_{c,r}\|$ is the Rao score statistic for a small dent at (c, r) against H_0. We evaluate it for all candidate centres and two radii with a sparse matrix product, and start H_1 fits from the top-k after non-maximum suppression. Proposals are computed from the nuisance-only residual, and each start is fitted both with and without the H_0 albedo patch, so a colour explanation fitted first cannot hide a dent.

Decision. The fit gain is $\Delta = L_\Omega(H_0) - L_\Omega(H_1) \geq 0$, as H_1 nests H_0. We report D if $\Delta > \tau$ and $D = 0$ otherwise, with τ the split-conformal $(1-\alpha)$ quantile of Δ on calibration negatives (healthy and colour-only blades), which bounds the false-change rate on exchangeable negatives by α. We use a Charbonnier loss ($\epsilon = 2/255$) summed over Ω; a Student-t noise model and a wider Charbonnier floor did not improve separation on development data.

4. The BladeDelta Benchmark

Geometry. Healthy blades are public Rotor37 compressor geometries [5] (CC BY-SA 4.0), normalised to unit span and closed with a labelled artificial hub collar. Each mesh has 29,989 vertices. Shapes 0–12 were used during earlier development and only to train the shader. Shapes 13–24 form the development split, 25–36 the calibration split and 37–66 the sealed test split. Shapes 67–96 form a second sealed test split, rendered only after the post-hoc variants of Section 5 were frozen; 97–109 remain in reserve.

Damage. Each blade receives one of: no change, a colour-only patch (a soft darkened or rust-tinted albedo change), a dent, or a dent plus a colour patch (half overlapping). Dents follow one of three families — a cubic profile with a raised rim, a Gaussian, and an anisotropic elliptic profile (aspect 0.4–1) — with depth log-uniform in $[0.0005, 0.006]$ blade spans (about 0.04–0.45 mm on a 74 mm Rotor37 blade) and radius in $[0.04, 0.12]$. Centres face at least one of seven cameras and lie at least $\max(0.1, 1.2r)$ from the airfoil boundary.

Capture. Images are rendered with Blender Cycles (32 samples, denoised, AgX tone mapping, 512×512) from seven cameras on a ring around the blade, under one of four conditions: nominal; camera jitter (yaw, pitch and roll each uniform in $\pm 1.5°$ per view); light jitter (each of three area lights $\times[0.85, 1.15]$); or both. Methods receive only the nominal cameras and lighting. The default input is two views (60° and 120°); four- and seven-view settings are ablations.

Protocol. The split, case lists, metrics and hypotheses were hash-frozen before any test image was rendered;

Split	Blades	Healthy	Stain	Dent	Dent + stain
Shader training	13	only for the renderer; never evaluated			
Development	12	12	24	36	24
Calibration	12	24	36	36	24
Test (sealed)	30	30	60	90	60
Test 2 (sealed)	30	30	60	90	60

Table 1. BladeDelta case counts. Each case has 7 rendered views; methods use 2 by default.

True depth (mm)	2 views	2 views (same cases)	4 views	7 views
< 0.1	1/10 (1 loc.)	0/4 (0 loc.)	1/5 (0 loc.)	3/9 (3 loc.)
0.1–0.2	7/14 (7 loc.)	1/2 (1 loc.)	2/6 (2 loc.)	7/11 (7 loc.)
> 0.2	12/13 (12 loc.)	4/4 (4 loc.)	4/6 (4 loc.)	8/9 (6 loc.)
All visible dents	20/37 (20 loc.)	5/10 (5 loc.)	7/17 (6 loc.)	18/29 (16 loc.)

Table 2. Visible test dents reported (located within 0.1 span) by true depth, at the frozen threshold. Below 0.1 mm a dent changes the image by about one grey level; the method declines rather than guess.

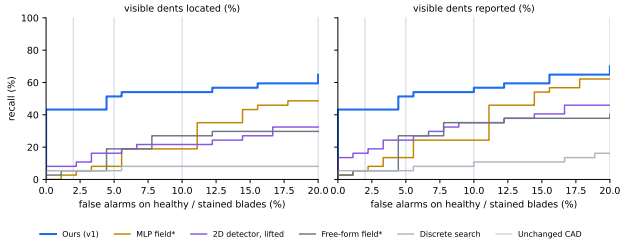


Figure 1. Recall of visible dents against the false-alarm rate on healthy and stain-only blades, as each method's threshold is swept on the sealed test (left: located within 0.1 span; right: reported anywhere). *Reimplementations of the core idea of CADSplat [20].

protocol changes are logged with dates. For each damaged case we record observability, the depth-weighted fraction of the dent visible in the input views.

CAD6 transfer split. To test transfer beyond Rotor37 we add six industrial blade designs supplied as STEP solids (airfoil spans 24–70 mm), with platforms, fillets and dovetail roots. The airfoil is the largest B-rep face (two faces on one blade); everything else is excluded from evidence like the Rotor37 collar. Each blade is normalised to unit airfoil span in the Rotor37 frame and receives 76 cases (456 in total): 30% dents centred where both input cameras see the surface, 15% dents hidden from both, 20% healthy, 15% stain only, 10% dent plus stain and 10% healthy with camera error alone ($\pm 0.5°$, $\pm 1°$, $\pm 1.5°$). Dents are sized in millimetres as an inspector would specify them: depth log-uniform in $[0.1, 1.0]$ mm and radius in $[1, 5]$ mm. The case list was hashed before rendering, and no component — shader, threshold or hyperparameter — was refit on this split.

Metrics. Relative local error is the RMS vertex error over the true dent support divided by that of the unchanged CAD (1: no better than CAD; 0: exact). We also report relative whole-airfoil error, depth error, centre error, detection, false physical change on negatives (any vertex moved more than 2×10^{-5}) and abstention on unobservable dents, with 95% bootstrap intervals resampled by blade shape.

5. Experiments

Baselines. Unchanged CAD reports the healthy mesh. Discrete search re-implements our earlier method: image-guided proposals, a fixed cubic profile of radius 0.08 and depths $\{0.0005, 0.001, 0.0015\}$, at nominal pose. Free-form field fits a per-vertex normal displacement with Laplacian smoothness and ℓ_1 sparsity after the same pose and lighting fit. MLP deformation field replaces the per-vertex values with a coordinate MLP (positional encoding, two hidden layers), the continuous deformation field used by CAD-prior reconstruction. These two baselines reimplement the core idea of CADSplat [20] — a smooth deformation field fit to sparse posed views under a CAD prior — within our renderer; they are not the official model, whose code had not been released at the time of writing. 2D detector, lifted is a U-Net trained on (photo, healthy render) pairs of separate exposed blades to segment dents; detections are back-projected onto the mesh through the nominal cameras. All methods share the renderer, cameras and observations, and each is thresholded by the same calibration procedure on its own score.

Main results. On the sealed test the method reports 12 of the 13 visible dents deeper than 0.2 mm, each at the correct location, and declines most dents below 0.1 mm, where the image change is about one grey level (Table 2). Reported dents are measured to a median depth error of 0.03 mm and centre error of 0.13 mm (Table 3).

Comparison at matched false-alarm rates. Each method in Table 3 uses its own calibrated threshold, so its recall and false-alarm rate are not directly comparable. Figure 1 and Table 4 remove this: at every false-alarm rate from 2% to 20%, the method locates more visible dents than any baseline (43% at 2% false alarms, 51% at 5%), while the strongest baselines locate 8–19% at 5%. The CADSplat-style fields catch

Method	AUROC ↑	Found % ↑	Located % ↑	False alarm % ↓	Abstain (hidden) % ↑	Depth err. (mm) ↓	Centre err. (mm) ↓
Ours (v1, frozen)	0.82 [0.71, 0.90]	54 [39, 68]	54 [39, 68]	10 [4, 17]	85 [77, 91]	0.03	0.13
without the evidence test	0.82 [0.71, 0.90]	84 [68, 95]	70 [56, 84]	32 [21, 43]	45 [36, 54]	0.03	0.68
2D detector, lifted	0.71 [0.63, 0.80]	24 [12, 40]	16 [5, 30]	4 [1, 9]	96 [93, 99]	–	1.82
MLP deformation field†	0.74 [0.65, 0.81]	16 [6, 29]	11 [3, 22]	6 [1, 10]	100 [100, 100]	–	1.58
Free-form field†	0.63 [0.54, 0.70]	14 [4, 24]	11 [3, 20]	4 [1, 9]	99 [97, 100]	–	4.69
Discrete search	0.53 [0.43, 0.65]	5 [0, 13]	5 [0, 13]	6 [1, 10]	98 [95, 100]	–	1.60
Unchanged CAD	0.50 [0.50, 0.50]	0 [0, 0]	0 [0, 0]	0 [0, 0]	100 [100, 100]	–	–

Table 3. Sealed test (240 cases, 30 unseen Rotor37 blades). Found / located: visible dents (≥50% visible in the two input views) reported / reported within 0.1 span of the true centre. Each method uses its own threshold, calibrated on the calibration split (split-conformal, $\alpha = 0.05$); "without the evidence test" reports every fit. Errors are medians over reported dents. Mean [95% interval, resampling blades]. †Reimplementations of the core idea of CADSplat [20], not the official model.

Method	@2%	@5%	@10%	@20%
Ours (v1)	43 (43)	51 (51)	54 (57)	65 (70)
MLP field†	3 (5)	8 (14)	19 (24)	49 (62)
2D detector, lifted	8 (16)	16 (24)	22 (35)	32 (46)
Free-form field†	5 (5)	19 (27)	27 (35)	30 (41)
Discrete search	5 (5)	5 (5)	8 (11)	8 (16)
Unchanged CAD	0 (0)	0 (0)	0 (0)	0 (0)

Table 4. Recall at matched false-alarm rates on the sealed test: visible dents located within 0.1 span (in brackets: reported anywhere), with each method's threshold placed so that at most 2/5/10/20% of the 90 healthy and stain-only cases raise a claim (37 visible dents). This compares score rankings at equal false alarms. †Reimplementations of the core idea of CADSplat [20].

Operating point	α	τ	Located %	False alarm %	Abstain (hidden) %	Use
Conservative	0.02	1.71	54	10	87	safety-critical sign-off
Balanced (paper)	0.05	1.52	54	10	85	routine inspection
Screening	0.20	0.45	68	22	61	triage, human review

Table 5. Three deployable operating points of the same frozen model. Each threshold is the split-conformal quantile of fit gains on the calibration split at level α; numbers are on the sealed test. With 60 calibration negatives, $\alpha = 0.02$ is the strictest certifiable level.

Case	n	Dent claimed	of which located	Declined ("not supported")
Healthy	30	1	–	29
Stain only	60	8	–	52
Visible dent	37	20	20	17
Hidden dent	113	17	11	96

Table 6. Every sealed-test decision of the frozen model ($\tau = 1.52$). A false alarm is a claim on a healthy or stain-only case; an abstention is a decline on a dent: correct when the dent is hidden, a miss when it is visible. The two never overlap.

up only at high false-alarm rates, because they explain stains as shape and must be thresholded above them.

Operating points and the meaning of an abstention. The same frozen model serves different needs by changing only the calibration level (Table 5); a screening setting ($\alpha = 0.2$) locates 68% of visible dents at 22% false alarms. Table 6 lists every decision: false alarms are claims on healthy or stain-only blades (9 of 90), and abstentions are declines on dents, correct for hidden dents and misses for visible ones; the two are disjoint.

Decision rule	Target FA (%)	Actual FA (%)	Visible dents found (%)
Ours: conformal on 60 calibration negatives	5	10	54
Ours: conformal on 360 calibration negatives	2	4	46
Textbook χ^2_8 likelihood-ratio threshold	5	32	70
Textbook χ^2_8 likelihood-ratio threshold	1	32	70
A-contrario detector on the H_0 residual (NFA < 1)	≈0	96	100

Table 7. Calibrated versus textbook decision rules on the same sealed-test fits. FA: claims on the 90 healthy or stain-only cases. Found: visible dents located (for the a-contrario detector, any detection). The textbook rules assume i.i.d. image noise; the real nuisance mismatch is far larger, so they cannot control false alarms.

Why not a standard statistical test? The fit gain is a likelihood ratio, so a textbook threshold might seem enough. It is not (Table 7). A χ^2_8 threshold derived from the per-view residual noise, asked for 1% false alarms, raises a false alarm on 32% of healthy and stain-only test cases, about thirty times its target, and the 5% setting gives the same 32%. An a-contrario detector on the nuisance-only residual [9], which controls the number of false alarms under an i.i.d. noise model, flags 96% of them. Both assume ideal image noise, while the real residual after the best nuisance explanation is dominated by structured mismatch: residual pose error, lighting, and the gap between the learned shader and the renderer. The split-conformal threshold is set on that real mismatch and stays near its target (10% at a nominal 5% with 60 calibration negatives; 4% at 2% with 360). Calibration is therefore not a cosmetic step: it is what makes false-alarm control hold under realistic image conditions.

Views	Observable dents	AUROC ↑	Located % ↑	Median rel. local err. ↓
2 views	10/60	0.847	60	0.43
4 views	17/60	0.777	41	1.00
7 views	29/60	0.857	62	0.47

Table 8. More views make more dents observable and the evidence clearer.

Variant	AUROC ↑	Located % ↑
Full method	0.855	72
No camera (pose) correction	0.679	72
No stain (colour) model	0.753	83
Whole image instead of interior evidence	0.685	61
No score-test proposals	0.420	6
No low-frequency brightness field	0.846	72

Table 9. Ablations on the development subset (18 observable dents, 18 healthy or stained blades).

Predicting detectability. The same nuisance-only fit predicts, before any search, how much evidence a dent of a given shape would produce once pose, lighting and a stain at the same place have explained what they can (the dent's information after profiling out the nuisance). On the actual captures this prediction separates dents the method finds from those it misses with AUROC 0.90 on the sealed test and 0.95 on the second sealed test, against 0.85 for the best depth-and-visibility heuristic. Aggregated into per-point maps of the minimum detectable depth for a single nominal capture, the same quantity is optimistic for shallow dents: it implies 49–66% recall below 0.1 mm and 93% at 0.1–0.2 mm, where the method achieves 10% and 50%; the maps assume the search reaches the right location and use a local linearisation, so we report them as an upper bound, not a calibrated limit.

Ablations.

Why recall is limited. Recall is set by dent depth and by the false-alarm budget, not by a failure to locate (Table 2). Every reported dent is in the right place, and 12/13 visible dents deeper than 0.2 mm are reported; most misses are shallower than 0.1 mm, where the change is about one grey level. In 6 of the 17 misses the dent hypothesis converged on the true location, but its evidence stayed below $\tau = 1.52$. The threshold is held up by appearance changes: stains account for 8 of the 9 false alarms, with stain-only fit gains up to 4.7 (5.6 for one healthy blade under camera error). Lowering τ to 0.54 would reach 70% recall at a 20% false-alarm rate. The misses are therefore the price of the false-alarm guarantee: an inspector receives few spurious damage claims, and dents below about 0.1 mm are reported as unsupported rather than guessed.

Method	AUROC	Found (%)	Located (%)	False alarm (%)	Abstain hidden (%)
Ours, Rotor37 sealed test	0.82 [0.71, 0.90]	54 [39, 68]	54 [39, 68]	10 [4, 17]	85 [77, 91]
Ours	0.74 [0.67, 0.84]	94 [89, 98]	70 [60, 81]	67 [36, 97]	33 [3, 67]
2D detector, lifted	0.86 [0.83, 0.89]	42 [35, 49]	36 [30, 42]	6 [3, 8]	100 [100, 100]
MLP deformation field	0.73 [0.68, 0.81]	84 [70, 97]	54 [47, 62]	43 [27, 58]	67 [48, 85]
Free-form field	0.60 [0.56, 0.66]	61 [36, 84]	37 [21, 53]	39 [20, 58]	64 [48, 82]
Discrete search	0.65 [0.58, 0.72]	16 [12, 18]	13 [10, 17]	4 [0, 10]	97 [94, 100]
Unchanged CAD	0.50 [0.50, 0.50]	0 [0, 0]	0 [0, 0]	0 [0, 0]	100 [100, 100]
Ours v1 (frozen), held-out blades 3–6	0.72 [0.65, 0.90]	90 [85, 95]	67 [55, 79]	51 [10, 91]	50 [9, 91]
Ours v1.1, held-out blades 3–6	0.95 [0.93, 0.98]	86 [77, 95]	83 [76, 90]	4 [0, 11]	98 [93, 100]
Ours, CAD6 leave-one-blade-out τ	0.70 [0.63, 0.84]	13 [3, 26]	6 [1, 13]	4 [0, 11]	95 [89, 100]

Table 10. Transfer to six unseen real blade designs (CAD6, 456 cases; dents 0.1–1.0 mm deep). Every method is frozen: shader trained on Rotor37 only, thresholds from the Rotor37 calibration split. Top row: ours on the Rotor37 sealed test, for the transfer gap. Brackets: 95% intervals resampling blades.

Blade	Airfoil span (mm)	Cases	Visible dents	AUROC	Found (%)	False alarm (%)
1	24	76	31	0.98	100	100
2	26	76	31	0.72	100	100
3	31	76	31	0.64	97	100
4	40	76	31	0.66	94	82
5	52	76	31	0.93	84	6
6	70	76	31	0.86	87	15

Table 11. Ours on each CAD6 blade, smallest to largest airfoil (frozen threshold).

Transfer to unseen blade designs. Table 10 applies every method, unchanged, to the six CAD6 blades (456 cases). Separation falls from 0.82 on the Rotor37 sealed test to 0.74 (transfer gap 0.08), with 94% of visible dents found, 67% false alarms and 33% abstention on hidden dents. The threshold was calibrated on Rotor37 only, so the false-alarm rate here measures how far the guarantee survives a change of geometry, not a property the method was fit to. Per blade (Table 11), dents are still ranked above negatives (best AUROC on blade 1 (24 mm, AUROC 0.98), worst on blade 3 (31 mm, AUROC 0.64)), but the frozen threshold fails on the smaller blades: false alarms fall with airfoil span (Spearman $\rho = -0.88$, $p = 0.02$). Across blades, airfoil span and detection rate have Spearman $\rho = -0.93$ ($p = 0.01$; six blades, so weak evidence): 99% of visible dents were found on the three smaller blades and 88% on the three larger ones. Because dents are specified in millimetres, the same dent is relatively deeper on a small blade; detection tracks depth relative to the airfoil span ($\rho = 0.10$) more strongly than depth in millimetres ($\rho = 0.02$), rising from 92% in the shallowest relative-depth tertile to 97% in the deepest. Diagnosis and fix (v1.1). On a healthy CAD6 capture whose image is identical to the reference, the silhouette pose fit moves a correct camera by about 0.3° and a spurious dent absorbs the misregistration; under real camera error, the image-space reference bias — large on unseen geometry because the shader was trained on Rotor37 only — stays fixed while the blade moves. v1.1 corrects both: it applies the reference correction to the silhou-

Variant (CAD6, 48 cases)	AUROC	Found (%)	False alarm (%)
Full method	0.71 [0.62, 0.83]	92 [83, 100]	78 [56, 100]
No camera correction	0.90 [0.81, 0.98]	92 [83, 100]	56 [33, 67]
No stain model	0.73 [0.64, 0.84]	96 [88, 100]	89 [78, 100]
Random starts	0.45 [0.37, 0.56]	38 [29, 46]	44 [22, 67]
Fixed grid of starts	0.50 [0.41, 0.62]	33 [17, 54]	44 [11, 78]
Whole image	0.56 [0.44, 0.71]	75 [58, 92]	61 [50, 67]

Table 12. Ablations on CAD6 (frozen settings, fixed 48-case subset).

Method	AUROC	Found (%)	Located (%)	False alarm (%)	Abstain hidden (%)
v1 (frozen)	0.79 [0.68, 0.88]	54 [40, 69]	46 [32, 60]	6 [1, 11]	88 [81, 93]
v1.1 (camera-check + correction fix)	0.80 [0.70, 0.88]	52 [38, 67]	46 [32, 61]	8 [2, 14]	88 [81, 93]

Table 13. Second sealed test: 30 further Rotor37 blades (67–96), 240 cases, unused by any earlier experiment. Each method ran once with its frozen threshold.

ette as well as to shading, and it reprojects the shading correction through the surface so that it moves with the blade (identical to v1 at the nominal pose). Both changes were developed on CAD6 blades 1–2 only and frozen, with a threshold recalibrated on the Rotor37 calibration split, before blades 3–6 were run. On those held-out blades v1.1 raises AUROC from 0.72 to 0.95, cuts false alarms from 51% to 4%, locates 83% of visible dents (v1: 67%) and abstains on 98% of hidden dents (v1: 50%). Recalibrating the threshold on CAD6 negatives, leave-one-blade-out, gives 13% found at 4% false alarms: the cost of transferring a threshold across blade designs.

A pre-registered negative result (v2). Targeting the remaining false alarms, we pre-registered three changes and a rule on development blades: a smooth surface appearance field shared by both hypotheses (A), a dent-size prior (D), or both (E), adopted only if recall at matched false-alarm rate improved by at least five points. None did: D matched v1 (67% recall, AUROC 0.87 vs. 0.85), while A and E raised negative fit gains so much that recall fell to 6%; the extra appearance freedom helped the dent hypothesis as much as the null. The change that does reduce false alarms is v1.1's reference correction of silhouette and shading (above), which also lowers the worst development false-alarm gains on Rotor37 (2.46 to 0.09).

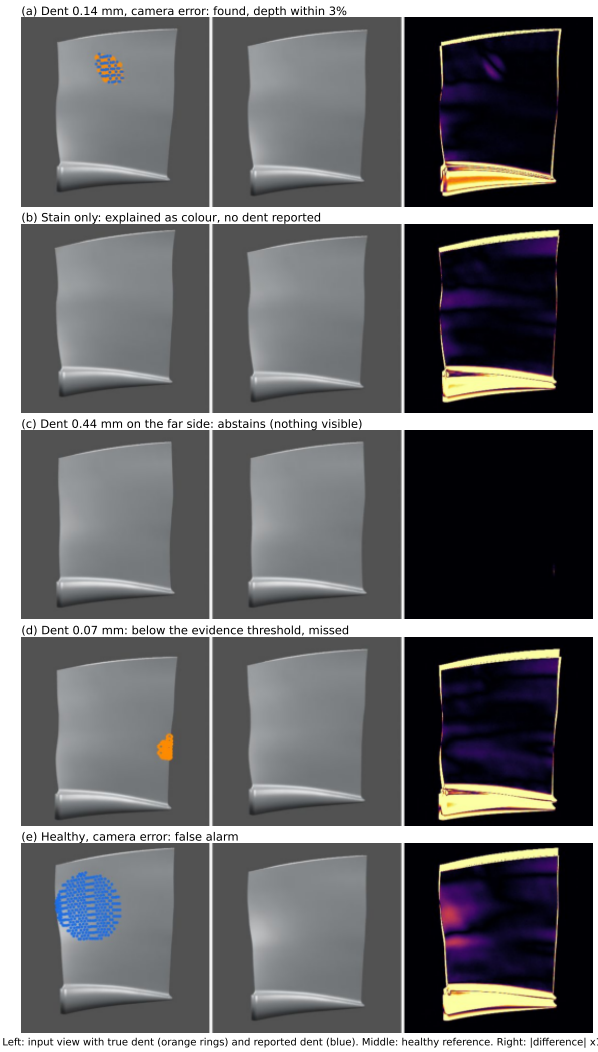


Figure 2. Sealed-test outcomes. (a) A 0.14 mm dent under camera error is found and measured. (b) A stain is explained as an albedo change. (c) A dent facing away from both cameras yields no evidence and the method abstains. (d) A 0.07 mm dent stays below the threshold. (e) A healthy blade under camera error is reported as dented, the residual failure mode.

Physical validation (pending real validation).

Pending real validation. A 2× print of CAD6 blade 5 will be dented with steel balls (target depths 0.3, 0.7 and 1.2 mm), photographed with marker-based poses before and after denting, and measured by structured-light scan (or a 0.01 mm dial gauge). This section is reserved for those results; the rest of the paper does not depend on it.

Qualitative results and failures. Figure 2 shows typical outcomes, including a miss and a false alarm.

Objects / defects / negatives	6 / 22 / 31
AUROC (defect vs. undamaged)	0.540
Detected (leave-one-object-out)	0/22
Located within 15 mm	0/22
False alarms	1/31

Table 14. Real objects (SiM3D) scanned before and after damage; truth from the measured scans.

Real objects. No public dataset pairs real blade photographs with measured damage geometry, so we use SiM3D [8]: six wooden stools scanned with an industrial structured-light sensor before and after damage, with calibrated photographs and expert defect annotations. The healthy scan replaces the CAD model, the healthy photograph from the same scanner position serves as the reference after 2D alignment, and the measured scan difference is the truth; annotated regions with no measured shape change (paint, contamination) count as negatives. On these textured objects our method does not separate physical defects from undamaged areas (Table 14). The residual between sessions is dominated by wood-grain misregistration and specular changes that the shader, trained on 12 photographs per object, does not model. We report this as a failure case: the evidence test is only as reliable as the appearance model behind it.

6. Limitations and Conclusion

Our positive results use rendered images of public compressor geometry with synthetic damage. On real scanned objects (SiM3D) the method did not separate physical defects from undamaged areas; real-blade validation with measured damage remains future work, and an appearance model that holds on real materials is its main prerequisite. The dent model has one dent per blade, and edge damage such as nicks and material loss is out of scope. The conformal guarantee holds for negatives exchangeable with the calibration set; a new camera rig or lighting setup requires recalibration. The method answers whether these views support a physical change; it does not certify airworthiness or replace an inspector.

We framed blade damage estimation as a decision with a stated level of evidence: report a dent, with its location and depth, only when a dent explains the images significantly better than every non-physical explanation, and abstain otherwise. On rendered Rotor37 blades this cuts the false-claim rate of an unconstrained fit by two thirds (32% to 10%) while keeping most detectable dents, and it outperforms CAD-prior deformation fields and a lifted 2D detector, whose lack of an appearance model forces their thresholds so high that few dents survive. Recall is bounded by what the cameras see: almost all misses are shallower than 0.1–0.2 mm, and seven views instead of two locate nearly three times as many dents at a lower false-alarm rate. Transfer to new blade designs exposed two assumptions that only held for the training family; making the healthy reference follow the silhouette and the camera removed the failure without retraining and without cost on Rotor37. Physical validation on a printed, dented and scanned blade (Section 5) is the next step.

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